

Co-author declaration for the following joint paper:

This declaration should describe the research contribution of the candidate, the main supervisor (where he/she is an associate author) and the other two most central authors (the corresponding author must be among them). If applicable, the contributions from other PhD candidates who have or intend to include the paper in a thesis should be described. Contributions from master students should be described.

Authors: *Ieva Rauluseviciute, Timothée Launay, Guido Barzaghi, Sarvesh Nikumbh, Boris Lenhard, Arnaud Regis Krebs, Jaime A. Castro-Mondragon and Anthony Mathelier*

Title: *Identification of transcription factor co-binding patterns with non-negative matrix factorization*

Journal: *in peer review at the date of thesis submission*

Ieva Rauluseviciute's independent contribution:

Co-first author, PhD candidate

COBIND method development and implementation. Design and implementation of all data analyses: data pre-processing, discovery of co-binding patterns and interpretation, analysis of evolutionary conservation, DNase I hypersensitivity data mining, processing, and analysis, single-molecule footprinting (SMF) data analysis. Collection, visualization, summarization, and interpretation of all results. Preparation of the COBIND tool and development of a web resource with all the co-binding patterns. Writing the first draft of all sections in the manuscript and supplementary material.

Guido Barzaghi

Co-author

Providing SMF data and its analysis package. Adaptation of the SMF analysis package for COBIND data analysis. SMF analysis results interpretation. Revision of the manuscript.

Jaime A. Castro-Mondragon

Co-author

Study design. Development and update of matrix clustering software. Contribution to the co-binding analysis results visualization and results interpretation. Revision of the manuscript.

Anthony Mathelier

Main supervisor, Corresponding author

Study design, supervision, and project management. Interpretation of the results. Coordination with international collaborators. Revision of the manuscript.

Has this paper been, or will this paper be part of another doctoral degree thesis?

Yes: No: **X**

If yes, elaborate:

Contributions from master students: None

Do you verify that **Ieva Rauluseviciute** has contributed to this joint paper as described above?

Yes: **X** No:

If no, specify:

Co-author's signatures:



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